Fiscal Year:	FY 2014	**Task Last Updated:**	FY 04/07/2014
PI Name:	Pandey, Santosh Ph.D.		
Project Title:	Integrative Response of C. elegans to Environmental Stresses in Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology (2) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology (2) Immunology		
Space Biology Special Category:	None		
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Zip Code:	50011-0001	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Space Biology NNH11ZTT002N
Start Date:	08/15/2012	**End Date:**	12/31/2013
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
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Flight Program:			
Flight Assignment:	NOTE: Extended to 12/31/2013 per A. Chu/ARC and NSSC information (Ed., 6/17/13)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Phillips, Gregory (Iowa State University) Powell-Coffman, Jo (Iowa State University) Vaidya, Umesh (Iowa State University)		
Grant/Contract No.:	NNX12AO60G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The proposed research aims to study the role of altered gravity on C. elegans in the presence of environmental stresses. Specifically, we will focus on the effects of hypoxia (i.e. oxygen deficiency) and pathogenesis on C. elegans exposed to hyper- or hypo-gravity situations. Previous studies using C. elegans have identified gravity-affected genes, measured muscle degradation, established spaceflight culture platforms, and developed a remote imaging system for worms. We propose to build on these previous studies supported by NASA and expand our fundamental understanding on the topic. Three objectives will be accomplished in the proposal: (i) Design microscale fluidic devices combined with real-time imaging to characterize worm behavior under stresses, (ii) Develop genetic and molecular techniques to study muscle degradation induced by hypoxia, and test whether/how this is affected by exposure to altered gravity, and (iii) Develop C. elegans pathogenesis models and identify key changes in host-pathogen relationships upon exposure to altered gravity. Collectively, the proposed research will help us understand the interplay between aging, environment, and infection on C. elegans, and lead to improved treatments and therapies to enhance resistance to bacterial pathogens during the stress conditions of spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research furthers the understanding of how hyper-gravity affects the biology of C. elegans organisms. We initially found that these worms can survive extremely high exposure to hyper-gravity (upto 500 G) in laboratory environments. Puzzled by this finding, we conducted rigorous tests by varying the duration of hyper-gravity exposure and the duration of recovery after the hyper-gravity exposure. A number of biological and biophysical parameters were characterized to verify whether the hyper-gravity stress produces any change in the worms. Such a detailed study was not available in the published literature. We found that some levels of significant changes in behavior were observed after 12 hours of hyper-gravity exposure but this effect was not seen at smaller or longer time durations. Infectivity of the wild-type worms were tested on three strains of bacteria and the worm survivability was assessed for over two weeks. In summary, we developed a range of measurable parameters to assess the effects of hyper-gravity on free-moving worms. We found that worm motility remains virtually unaltered over a wide range of hyper-gravity exposure (1 G to 500 G) for time durations as long as 24 hours. Only under 12 hours of exposure, worm motility was decreased to a noticeable range. We developed a microfluidic chip technology with real-time imaging to process worm videos automatically and extract the motility-related parameters. We are also developing a scanner-based imaging technology to count worm survivability for infection assays that will circumvent the use of manual labor.
Task Progress:	1) Microfluidic device design, fabrication, and worm phenotyping: We developed different microfluidic chip designs to measure the motility of free-moving C. elegans by recording real-time videos. An image tracking program was built to track the moving worms from the recorded videos and extract motility parameters such as worm velocity, number of stops, and behavior of a worm population. Our results on testing worm motility in microfluidics has been published in a top-tier peer-reviewed journal: Toxicological Sciences. 2) Methods to assay the effects of altered gravity: Through systematic and carefully controlled studies, we have assayed the effects of various C. elegans culture conditions on survival in hypergravity. We assayed varied times of exposure and recovery. Most wild-type C. elegans survived hyper gravity conditions, but survival was influenced by the media used. 3) Methods to assay the effects of altered gravity on C. elegans susceptibility to bacterial pathogens: We completed initial experiments establishing the methods for Salmonella pathogenesis studies. We also purchased a biosafety cabinet to allow pathogenesis studies proximal to the hypergravity studies. To enable frequent and safe monitoring of plates containing pathogenic bacteria, we are developing a scanner-based system to image the movement of worms on multiple agarose plates without human intervention for long time periods. Journal Publication: Jenifer Saldanha, Archana Parashar, Santosh Pandey, Jo Anne Powell-Coffman. "Multi-parameter behavioral analyses provide insights to mechanisms of cyanide resistance in Caenorhabditis elegans" Toxicological Sciences, June 2013, Impact Factor: 4.328
Bibliography Type:	Description: (Last Updated: 08/24/2017)
Articles in Peer-reviewed Journals	Saldanha JN, Parashar A, Pandey S, Powell-Coffman JA. "Multiparameter behavioral analyses provide insights to mechanisms of cyanide resistance in Caenorhabditis elegans." Toxicological Sciences. 2013 Sep;135(1):156-68. Epub 2013 Jun 26. http://dx.doi.org/10.1093/toxsci/kft138 ; PubMed PMID: 23805000; PubMed Central PMCID: PMC3748764 , Sep-2013
Articles in Peer-reviewed Journals	Lycke R, Parashar A, Pandey S. "Microfluidics-enabled method to identify modes of Caenorhabditis elegans paralysis in four anthelmintics." Biomicrofluidics. 2013 Nov 6;7(6):64103. http://dx.doi.org/10.1063/1.4829777 eCollection 2013. PubMed PMID: 24396537; PubMed Central PMCID: PMC3838407 , Nov-2013
Articles in Peer-reviewed Journals	Saldanha JN, Pandey S, Powell-Coffman JA. "The effects of short-term hypergravity on Caenorhabditis elegans." Life Sci Space Res (Amst). 2016 Aug;10:38-46. https://doi.org/10.1016/j.lssr.2016.06.003 ; PubMed PMID: 27662786 , Aug-2016